



Linear Equations

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01

Linear Equation

Systems of Linear Equations

Definition

A system of m linear equations with n unknowns:

- F is a field, we want to find n scalars (elements of F) $x_1, \dots, x_n$ which satisfy the conditions: (A_{ij}, y_k are elements of F)

$$A_{11}x_1 + A_{12}x_2 + \dots + A_{1n}x_n = y_1$$

$$A_{21}x_1 + A_{22}x_2 + \dots + A_{2n}x_n = y_2$$

...

$$A_{m1}x_1 + A_{m2}x_2 + \dots + A_{mn}x_n = y_m$$

If $y_1 = y_2 = \dots = y_m = 0$, we say that the system is homogeneous.

A solution of this system of linear equations is vector $\begin{bmatrix} s_1 \\ \vdots \\ s_n \end{bmatrix}$ whose components satisfy

$$x_1 = s_1, \dots, x_n = s_n$$

Linear Equation (Geometric Interpretation and Intuition)

- Consider this simple system of equations,

$$x - 2y = 1$$

$$3x + 2y = 11$$

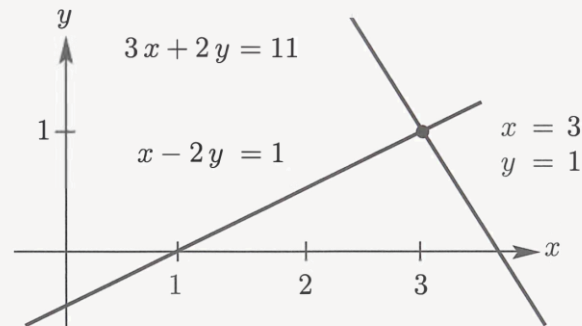
- Can be expressed as a matrix-vector multiplication

- Matrix Equation: $Ax=b$

$$\underbrace{\begin{bmatrix} 1 & -2 \\ 3 & 2 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x \\ y \end{bmatrix}}_{\vec{x}} = \underbrace{\begin{bmatrix} 1 \\ 11 \end{bmatrix}}_b$$

- A is often called **coefficient matrix** $\begin{bmatrix} 1 & -2 \\ 3 & 2 \end{bmatrix}$

- Ab is an **Augmented matrix**: $\begin{bmatrix} 1 & -2 & 1 \\ 3 & 2 & 11 \end{bmatrix}$



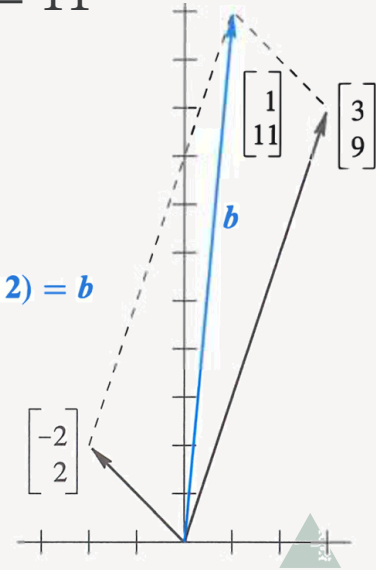
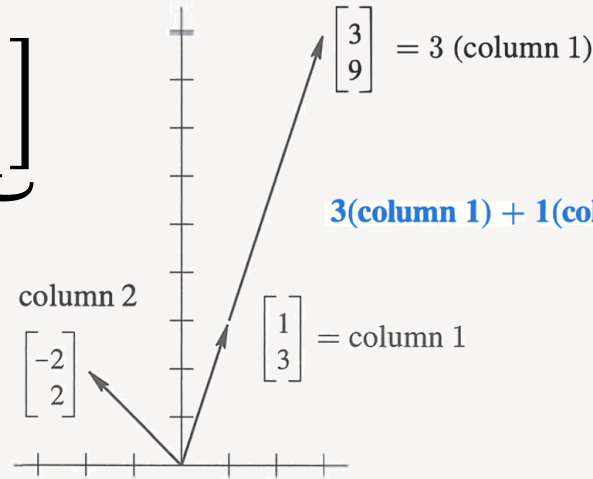
Vectors & Linear Equation

Also, Can be expressed as linear combination of cols:

$$\begin{aligned} x - 2y &= 1 \\ 3x + 2y &= 11 \end{aligned}$$

$$\underbrace{\begin{bmatrix} 1 & -2 \\ 3 & 2 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x \\ y \end{bmatrix}}_{\vec{x}} = \underbrace{\begin{bmatrix} 1 \\ 11 \end{bmatrix}}_b$$

$$x \begin{bmatrix} 1 \\ 3 \end{bmatrix} + y \begin{bmatrix} -2 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix} = b$$



Same for n equation, n variable

Idea Of Elimination

- Subtract a multiple of equation (1) from (2) to eliminate a variable

$$\begin{array}{r} x - 2y = 1 \\ 3x + 2y = 11 \end{array}$$

multiply equation 1 by 3
→
Subtract to eliminate 3x

$$\begin{array}{r} x - 2y = 1 \\ 8y = 8 \end{array}$$

$$\underbrace{\begin{bmatrix} 1 & -2 \\ 0 & 8 \end{bmatrix}}_U \underbrace{\begin{bmatrix} x \\ y \end{bmatrix}}_x = \underbrace{\begin{bmatrix} 1 \\ 8 \end{bmatrix}}_c$$

A has become an upper triangle matrix U

Existence and Uniqueness Questions

- A system of linear equations has:
 - No solution → inconsistent
 - Exactly one solution
 - Infinitely many solutions
- } → consistent

What we'll learn:

1. Is the system consistent? That is, does at least one solution exist?
2. If a solution exists, is it the only one? That is, is the solution unique?

02

Vector



Dot Product

Review & Geometric Interpretation

Categorical (Non-numerical) Data

- Sometimes you work with categorical data in machine learning.
- It is common to encode categorical variables to make them easier to work with and learn by some techniques. A popular encoding for categorical variables is the one hot encoding.
- A one hot encoding is:

Example (1)

$$\text{red} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{green} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\text{blue} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Categorical (Non-numerical) Data

- One-Hot Encodings (standard basis vector)
 - Assign to each word a vector with one 1 and 0s elsewhere.
 - Suppose our language only has four words:


$$\text{apple} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{cat} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{house} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\text{tiger} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$



Drawbacks

- ☐ Very sparse vectors.
- ☐ Are never similar!



How to measure the similarity?

- **Dot Product**

- The product of numbers is another number.
- The dot product of vectors is not another vector! It is a number!!


$$2 \times 5 = 10$$

Numbers

vs

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 5 \end{bmatrix} = 7$$

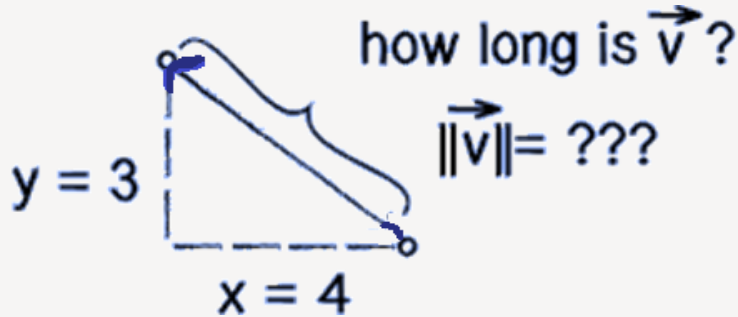
Vectors

A number

$$\begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} 7 \\ 2 \\ -1 \end{bmatrix} = (1)(7) + (0)(2) + (3)(-1) = 4$$

Length of vector

- Dot product between a vector and itself: magnitude-squared, the **length** squared, or the squared-norm, of the vector.



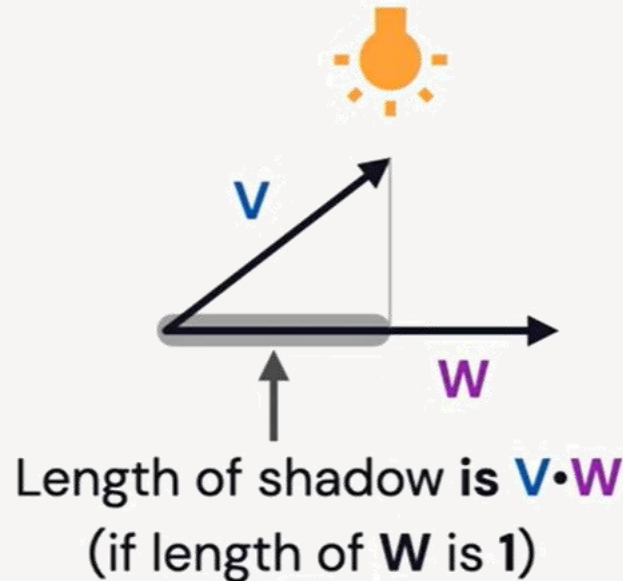
$$\mathbf{v} \cdot \mathbf{v} = \begin{bmatrix} 4 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 3 \end{bmatrix} = 16 + 9 = 25$$

Length($\mathbf{v}$) = 5

$$\mathbf{a}^T \mathbf{a} = \|\mathbf{a}\|^2 = \sum_{i=1}^n a_i a_i = \sum_{i=1}^n a_i^2$$

Dot Product (Geometric Interpretation and Intuition)

- Represents the length of the “shadow” of one vector along another.
- This indicates how similar the two vectors are.



One-Hot Encodings Drawbacks


$$\text{apple} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad \text{cat} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \quad \text{house} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \quad \text{tiger} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\text{apple} \cdot \text{cat} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} = \mathbf{0} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} = \text{tiger} \cdot \text{cat}$$


Vector Operations

- Vector-Vector Addition
- Vector-Vector Subtraction
- Scalar-Vector Product
- Vector-Vector Products:

- $\mathbf{x} \cdot \mathbf{y}$ is called the inner product or dot product or scalar product of the vectors: $\mathbf{x}^T \mathbf{y}$ or $\mathbf{y}^T \mathbf{x}$

■ $\langle a, b \rangle$ $\langle a | b \rangle$ (a, b) $a \cdot b$

$$\mathbf{x}^T \mathbf{y} \in \mathbb{R} = \begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \sum_{i=1}^n x_i y_i.$$

- Transpose of dot product:

■ $(a \cdot b) = a^T b = (a \cdot b)^T = (a^T b)^T = (b^T a) = (b \cdot a) = b^T a$

- Length of vector

03

Matrix Multiplication



Basic Notation

- By $A \in \mathbb{R}^{m \times n}$ we denote a matrix with m rows and n columns, where the entries of A are real numbers.

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} = \begin{bmatrix} | & | & \cdots & | \\ a_1 & a_2 & \cdots & a_n \\ | & | & \cdots & | \end{bmatrix} = \begin{bmatrix} - & a_1^T & - \\ - & a_2^T & - \\ & \vdots & \\ - & a_m^T & - \end{bmatrix}$$

Linear Combination

Definition

The **linear combinations** of m vectors $a_1, \dots, a_m$, each with size n is:

$$\beta_1 a_1 + \dots + \beta_m a_m$$

where $\beta_1, \dots, \beta_m$ are scalars and called the **coefficients of the linear combination**

Matrix-Vector Multiplication

- If we write A by rows, then we can express Ax as,


$$A \in \mathbb{R}^{m \times n} \quad y = Ax = \begin{bmatrix} - & a_1^T & - \\ - & a_2^T & - \\ & \vdots & \\ - & a_m^T & - \end{bmatrix} x = \begin{bmatrix} a_1^T x \\ a_2^T x \\ \vdots \\ a_m^T x \end{bmatrix}. \quad a_i^T x = \sum_{j=1}^n a_{ij} x_j$$

- If we write A by columns, then we have:

$$y = Ax = \begin{bmatrix} | & | & \cdots & | \\ a_1 & a_2 & \cdots & a_n \\ | & | & & | \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = [a_1]x_1 + [a_2]x_2 + \cdots + [a_n]x_n.$$

- y is a **linear combination** of the columns A .

We will learn in next lectures

columns of A are linearly independent if $Ax = 0$ implies $x = 0$



Matrix-Vector Multiplication

It is also possible to multiply on the left by a row vector.

- If we write A by columns, then we can express $x^T A$ as,


$$y^T = x^T A = x^T \begin{bmatrix} | & | & \cdots & | \\ a_1 & a_2 & \cdots & a_n \\ | & | & \cdots & | \end{bmatrix} = [x^T a_1 \quad x^T a_2 \quad \cdots \quad x^T a_n]$$

- Expressing A in terms of rows we have:

$$y^T = x^T A = [x_1 \quad x_2 \quad \cdots \quad x_m] \begin{bmatrix} - & a_1^T & - \\ - & a_2^T & - \\ & \vdots & \\ - & a_m^T & - \end{bmatrix}$$
$$= x_1 [- \quad a_1^T \quad -] + x_2 [- \quad a_2^T \quad -] + \dots + x_m [- \quad a_m^T \quad -]$$

- y^T is a linear combination of the rows of A.



Matrix-Vector Multiplication

- $A(u + v) = Au + Av$
- $(A + B)u = Au + Bu$
- $(\alpha A)u = \alpha(Au) = A(\alpha u) = \alpha Au$
- $0u = 0$
- $A0 = 0$
- $Iu = u$

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} = \begin{bmatrix} | & | & \cdots & | \\ a_1 & a_2 & \cdots & a_n \\ | & | & \cdots & | \end{bmatrix} = \begin{bmatrix} - & a_1^T & - \\ - & a_2^T & - \\ - & \vdots & - \\ - & a_m^T & - \end{bmatrix}$$

Example: Write in matrix-vector multiplication

- Column j : $a_j =$
- Row i : $a_i^T =$
- Vector sum of rows of $A =$
- Vector sum of columns of $A =$

$$A = \begin{bmatrix} -1 & 2 & 1 \\ 2 & 0 & -2 \end{bmatrix}$$

Matrix-Matrix Multiplication

Definition

Let A be an $m \times n$ matrix over the field F and let B be an $n \times p$ matrix over F . The product AB is the $m \times p$ matrix C whose i, j entry is:

$$C_{ij} = \sum_{r=1}^n A_{ir} B_{rj}$$

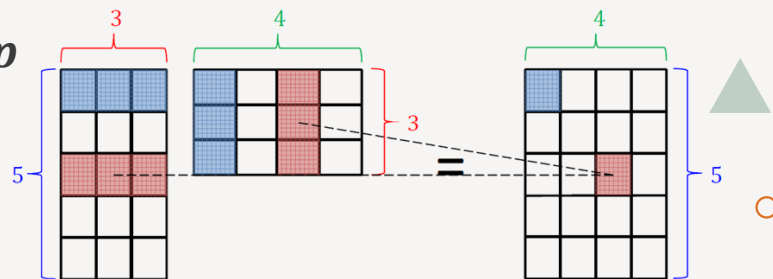
- $A \in \mathbb{R}^{m \times n}, B \in \mathbb{R}^{n \times p} \rightarrow \mathbb{R}^{m \times p}$
 - a_i rows of A , b_j cols of B

$$C = AB$$

for $1 \leq i \leq m, 1 \leq j \leq p$

dot product(a_i, b_j)

$$C_{ij} = a_i^T b_j$$



Matrix-Matrix Multiplication (different views)

1. As a set of vector-vector products


$$C = AB = \begin{bmatrix} - & a_1^T & - \\ - & a_2^T & - \\ & \vdots & \\ - & a_m^T & - \end{bmatrix} \begin{bmatrix} | & | & & | \\ b_1 & b_2 & \dots & b_p \\ | & | & & | \end{bmatrix} = \begin{bmatrix} a_1^T b_1 & a_1^T b_2 & \dots & a_1^T b_p \\ a_2^T b_1 & a_2^T b_2 & \dots & a_2^T b_p \\ \vdots & \vdots & \ddots & \vdots \\ a_m^T b_1 & a_m^T b_2 & \dots & a_m^T b_p \end{bmatrix}$$

2. As a sum of outer products

$$C = AB = \begin{bmatrix} | & | & & | \\ a_1 & a_2 & \dots & a_n \\ | & | & & | \end{bmatrix} \begin{bmatrix} - & b_1^T & - \\ - & b_2^T & - \\ & \vdots & \\ - & b_n^T & - \end{bmatrix} = \sum_{i=1}^n a_i b_i^T$$


Matrix-Matrix Multiplication (different views)

3. As a set of matrix-vector products.

$$C = AB = A \begin{bmatrix} | & | & \dots & | \\ b_1 & b_2 & \dots & b_p \\ | & | & \dots & | \end{bmatrix} = \begin{bmatrix} | & | & \dots & | \\ Ab_1 & Ab_2 & \dots & Ab_p \\ | & | & \dots & | \end{bmatrix}$$

Here the i th column of C is given by the matrix-vector product with the vector on the right, $c_i = Ab_i$. These matrix-vector products can in turn be interpreted using both viewpoints given in the previous subsection.

4. As a set of vector-matrix products.

$$C = AB = \begin{bmatrix} - & a_1^T & - \\ - & a_2^T & - \\ & \vdots & \\ - & a_m^T & - \end{bmatrix} B = \begin{bmatrix} - & a_1^T B & - \\ - & a_2^T B & - \\ & \vdots & \\ - & a_m^T B & - \end{bmatrix}$$

Matrix-Matrix Multiplication

- Properties:

- Associative

$$(AB)C = A(BC)$$

- Distributive

$$A(B + C) = AB + AC$$

- NOT commutative

$$AB \neq BA$$

- Dimensions may not even be conformable

04

Elementary Row Operations



Gaussian Elimination: Elementary Row Operations

- Elementary Row Operations
 1. **Scaling**: Multiply all entries in a row by a nonzero scalar.
 2. **Replacement**: Replace one row by the sum of itself and a multiple of another row.
 3. **Interchange**: Interchange two rows.
- Elementary Row Operation is a special type of function e on $m \times n$ matrix A and gives an $m \times n$ matrix $e(A)$ where $c \neq 0$.
 1. **Scaling**: $e(A)_{ij} = cA_{ij}$
 2. **Replacement**: $e(A)_{ij} = A_{ij} + cA_{kj}$
 3. **Interchange**: $e(A)_{ij} = A_{kj}$, $e(A)_{kj} = A_{ij}$

In defining $e(A)$, it is not really important how many columns A has, but the number of rows of A is crucial.

Inverse of Elementary Row Operation

Theorem (1)

The inverse operation (function) of an elementary row operation exists and is an elementary row operation of the same type.



Proof:

Proof. (1) Suppose e is the operation which multiplies the r th row of a matrix by the non-zero scalar c . Let e_1 be the operation which multiplies row r by c^{-1} . (2) Suppose e is the operation which replaces row r by row r plus c times row s , $r \neq s$. Let e_1 be the operation which replaces row r by row r plus $(-c)$ times row s . (3) If e interchanges rows r and s , let $e_1 = e$. In each of these three cases we clearly have $e_1(e(A)) = e(e_1(A)) = A$ for each A . ■



Row-Equivalent

Definition

If A and B are $m \times n$ matrices over the field F , we say that B is **row-equivalent** to A if B can be obtained from A by a finite sequence of elementary row operations.

Note (from Theorem 1 and the above definition)

- ❑ Each matrix is row-equivalent to itself
- ❑ If B is row-equivalent to A , then A is row-equivalent to B .
- ❑ If B is row-equivalent to A , C is row-equivalent to B , then C is row-equivalent to A

05

Elementary Matrices

Elementary Matrices

Definition

A $m \times m$ matrix is an elementary matrix if it can be obtained from the $m \times m$ identity matrix by means of a **single elementary row operation**.

Example

Find all 2×2 elementary matrices.

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \begin{bmatrix} 1 & c \\ 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 \\ c & 1 \end{bmatrix} \\ \begin{bmatrix} c & 0 \\ 0 & 1 \end{bmatrix}, \quad c \neq 0, \quad \begin{bmatrix} 1 & 0 \\ 0 & c \end{bmatrix}, \quad c \neq 0.$$

Elementary Matrices and Elementary Row Operation

Theorem (2)

Let e be an elementary row operation and let E be the $m \times m$ elementary matrix $E = e(I)$. Then, for every $m \times n$ matrix A :

$$e(A) = EA$$

Proof:

Proof. The point of the proof is that the entry in the i th row and j th column of the product matrix EA is obtained from the i th row of E and the j th column of A . The three types of elementary row operations should be taken up separately. We shall give a detailed proof for an operation of type (ii). The other two cases are even easier to handle than this one and will be left as exercises. Suppose $r \neq s$ and e is the operation 'replacement of row r by row r plus c times row s .' Then

$$E_{ik} = \begin{cases} \delta_{ik}, & i \neq r \\ \delta_{rk} + c\delta_{sk}, & i = r. \end{cases}$$

Therefore,

$$(EA)_{ij} = \sum_{k=1}^m E_{ik}A_{kj} = \begin{cases} A_{ij}, & i \neq r \\ A_{rj} + cA_{sj}, & i = r. \end{cases}$$

In other words $EA = e(A)$. ■

Multiplication of a matrix on the left by a square matrix performs row operations.

Elementary Matrices

Example

Matrix	Elementary row operation	Elementary matrix
$\begin{bmatrix} 1 & 0 & 2 \\ -2 & 0 & -3 \\ 0 & 2 & 0 \end{bmatrix}$	$R_2 \leftarrow R_2 + 2R_1$	$E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 0 & 1 \\ 0 & 2 & 0 \end{bmatrix}$	$R_2 \leftrightarrow R_3$	$E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$
$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$R_2 \leftarrow \frac{1}{2}R_2$	$E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$
$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$R_1 \leftarrow R_1 + (-2)R_3$	$E_4 = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$		

From property
 $(AB)C = A(BC)$

$$E_4(E_3(E_2(E_1A))) = (E_4(E_3(E_2E_1)))A$$

Row-Equivalent and Elementary Matrices

Theorem (3)

Let A and B be $m \times n$ matrices. Then B is row-equivalent to A if and only if $B = PA$, where P is a product of $m \times m$ elementary matrices.

Proof:

Corollary. Let A and B be $m \times n$ matrices over the field F . Then B is row-equivalent to A if and only if $B = PA$, where P is a product of $m \times m$ elementary matrices.

Proof. Suppose $B = PA$ where $P = E_s \cdots E_2 E_1$ and the E_i are $m \times m$ elementary matrices. Then $E_1 A$ is row-equivalent to A , and $E_2(E_1 A)$ is row-equivalent to $E_1 A$. So $E_2 E_1 A$ is row-equivalent to A ; and continuing in this way we see that $(E_s \cdots E_1)A$ is row-equivalent to A .

Now suppose that B is row-equivalent to A . Let $E_1, E_2, \dots, E_s$ be the elementary matrices corresponding to some sequence of elementary row operations which carries A into B . Then $B = (E_s \cdots E_1)A$. ■

Homogenous system

Example

Find the solution for this system.

Suppose F is the field of complex number and the coefficient matrix is:

$$A = \begin{bmatrix} -1 & i \\ -i & 3 \\ 1 & 2 \end{bmatrix}$$

In performing row operations it is often convenient to combine several operations of type (2). With this in mind

$$\begin{bmatrix} -1 & i \\ -i & 3 \\ 1 & 2 \end{bmatrix} \xrightarrow{(2)} \begin{bmatrix} 0 & 2+i \\ 0 & 3+2i \\ 1 & 2 \end{bmatrix} \xrightarrow{(1)} \begin{bmatrix} 0 & 1 \\ 0 & 3+2i \\ 1 & 2 \end{bmatrix} \xrightarrow{(2)} \begin{bmatrix} 0 & 1 \\ 0 & 0 \\ 1 & 0 \end{bmatrix}.$$

Thus the system of equations

$$\begin{aligned} -x_1 + ix_2 &= 0 \\ -ix_1 + 3x_2 &= 0 \\ x_1 + 2x_2 &= 0 \end{aligned}$$

has only the trivial solution $x_1 = x_2 = 0$.

Conclusion

- System of linear equations
- Different view of matrix multiplication
- Linear combination and matrix multiplication
- Associativity of three matrices multiplication
- Gaussian Elimination
- Row-equivalent of two matrices
- Elementary matrices



Resources

- ❑ Chapter 1: Kenneth Hoffman and Ray A. Kunze. Linear Algebra. PHI Learning, 2004.
- ❑ Chapter 1: David C. Lay, Steven R. Lay, and Judi J. McDonald. Linear Algebra and Its Applications. Pearson, 2016.
- ❑ Chapter 2: David Poole, Linear Algebra: A Modern Introduction. Cengage Learning, 2014.
- ❑ Chapter 1: Gilbert Strang. Introduction to Linear Algebra. Wellesley-Cambridge Press, 2016.

